

Hornsby Girls' High – 2006

Mathematics Preliminary Examination

Question 1 (14 marks)

- a) Evaluate the following. Express your answer in Scientific Notation correct to 3 significant figures. 2

1

$$\frac{-0.1}{\sqrt{(0.1)^2 + 8}}$$

- b) State the exact value of $\cos 210^\circ$ 2

- c) Show that the roots of the quadratic equation $3x^2 - 8x + 10 = 0$ are not real. 2

- d) Find $\lim_{x \rightarrow 3} \frac{x^2 - 9}{2x - 6}$ 2

- e) Simplify: $\tan \theta \times \sin (90^\circ - \theta)$ 2

- f) A ship sails due south for 10km and due west for 3km. Calculate its bearing from its starting point. 2

- g) Solve simultaneously: $2x + 5y = 16$
 $10x - 3y = -4$ 2

Question 2 (16 marks) **START A NEW PAGE**

- a) Differentiate the following:

i) $y = x^4 - 3x^3 + 12x$ 1

ii) $y = (3x + 5)(8 - 2x)$ 2

iii) $y = \frac{x}{x^2 + 4}$ 2

b) For the parabola $(x - 4)^2 = 12(y + 3)$ find:

- | | | |
|------|-------------------------------|---|
| i) | the vertex | 1 |
| ii) | the focus | 1 |
| iii) | the equation of its directrix | 1 |

c) For the quadratic $2x^2 - 6x + 1 = 0$ find:

- | | | |
|------|----------------------|---|
| i) | $\alpha + \beta$ | 1 |
| ii) | $\alpha\beta$ | 1 |
| iii) | $\alpha^2 + \beta^2$ | 2 |

d) Solve for θ : $0 \leq \theta \leq 360^\circ$
 $2 \sin \theta - \sqrt{3} = 0$ 2

e) Solve for x : $4^x - 9(2^x) + 8 = 0$ 2

Question 3 (15 marks) START A NEW PAGE

a) If $f''(x) = 6x - 8$, $f'(2) = 0$ & $f(-1) = 3$.
 Find an expression for $f(x)$. 2

b) Find the equation of a parabola with focus (5,-3)
 and directrix $x = 9$. 2

c) For the equation $y = \frac{1}{3}x^3 - x^2 - 8x - 6$ find:

- | | | |
|------|--|---|
| i) | the y-intercept | 1 |
| ii) | the stationary points and determine their nature | 2 |
| iii) | any inflection points | 2 |

Sketch the curve in the domain $-4 \leq x \leq 6$. 3
 State the minimum value of the function in the above domain. 1

d) Find the equation of the normal to the curve
 $y = \frac{1}{x^2}$ at the point where $x = 2$. Express your
 answer in general form. 2

Question 4 (14 marks) START A NEW PAGE

- a) What is the range of $y = 2\sqrt{9 - x^2}$? 2
- b) For what values of a is the line $y = ax$ a tangent to the hyperbola $y = \frac{3}{x - 2}$? 2
- c) Differentiate: $y = (\sqrt{x} + 1)^{100}$ 2
- d) Consider the function $f(x) = 2x^2 + x$
- i) Find $f'(x)$ using the First Principles Method 2
- ii) Show that $a(1 + f'(a)) = 2f(a)$. 2
- e) Show that the line $5x - 12y + 26 = 0$ is a tangent to the circle $x^2 + y^2 = 4$. 2
- f) Find the domain over which the curve $f(x) = x^3 - 7x^2 + 1$ is concave downwards. 2

Question 5 (13 marks) START A NEW PAGE

- a) What is the domain of $y = \sqrt{x^2 - 3x - 4}$? 2
- b) Sketch a curve which is monotonic increasing and concave up for all values of x . 2
- c) Prove: $\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \cos^2 \theta - \sin^2 \theta$ 3

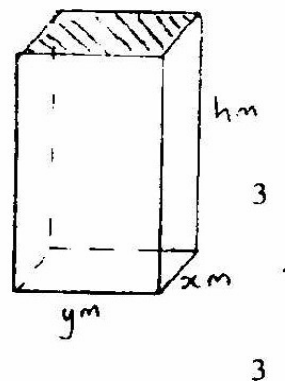
- d) A box with a rectangular base and height h metres is open at the top and has a volume of 32m^3 . The dimensions of the base of the box are x & y .

- i) Show that the surface area (A) of the box is

$$A = xy + 64\left(\frac{1}{x} + \frac{1}{y}\right)$$

- ii) Considering that for the surface area above y is constant and x is the variable, show that the minimum surface area of the

box is $A = \left(16\sqrt{y} + \frac{64}{y}\right)\text{m}^2$.



END OF PAPER

SOLUTIONS - YEAR 11 YEARLY EXAM 2006

11 a) -3.53×10^{-2}

i) $\cos 2\omega = -\cos 3\omega$
 $= -\frac{\sqrt{3}}{2}$

i) $3x^2 - 8x + 10 = 0$

$\Delta = b^2 - 4ac$

$= 64 - 4(3)(10)$

$= -56 < 0 \therefore \text{unreal roots}$

i) $\lim_{x \rightarrow 3} \frac{x^2 - 9}{2x - 6} \quad x \neq 3$

$= \lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{2(x-3)}$

$= 3$

i) $\tan \theta \times 2 \sin(90 - \theta)$

$= \frac{\sin \theta}{\cos \theta} \times \cos \theta$

$= \sin \theta$

i) $\tan \theta = \frac{3}{10}$

$\therefore \theta = 16^\circ 42'$

bearing is $196^\circ 42' T$

or $S 16^\circ 42' W$

i) $2x + 5y = 16 \quad \text{--- (1) } \times 5$

$10x - 3y = -4 \quad \text{--- (2)}$

$10x + 25y = 80 \quad \text{--- (3)}$

$-28y = -84$

$\therefore y = 3$

$2x + 15 = 16$

$\therefore x = \frac{1}{2}$

$\therefore x = \frac{1}{2}, y = 3$

12 a) i) $y = x^4 - 3x^3 + 12x$

$y' = 4x^3 - 9x^2 + 12$

i) $y = (3x+5)(8-2x)$

$y' = (8-2x)(3) + (3x+5)(-2)$

$y' = 24 - 6x - 6x - 10$

$y' = 14 - 12x$

Q2. (cont.)

iii) $y = \frac{x}{x^2 + 4}$

$y' = \frac{(x^2 + 4)(1) - x(2x)}{(x^2 + 4)^2}$

$\therefore y' = \frac{4 - x^2}{(x^2 + 4)^2}$

b) $(x-4)^2 = 12(y+3)$

$(x-4)^2 = 4(3)(y+3)$

i) $\therefore$ Vertex $(4, -3)$

ii) focus is $(4, 0)$

iii) directrix: $y = -6$

c) $2x^2 - 6x + 1 = 0$

i) $\alpha + \beta = 3$

ii) $\alpha\beta = \frac{1}{2}$

iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= 3^2 - 2(\frac{1}{2})$

$= 8$

d) $2\sin \theta - \sqrt{3} = 0$

$\sin \theta = \frac{\sqrt{3}}{2}$

$\frac{180 - 60}{2} = 60$

$\therefore \theta = 60^\circ, 120^\circ$

e) $4^x - 9(2^x) + 8 = 0$

let $m = 2^x$

$\therefore m^2 - 9m + 8 = 0$

$(m-8)(m-1) = 0$

$\therefore m = 8, m = 1$

but $m = 2^x$

$\therefore 2^x = 8, 2^x = 1$

$\therefore x = 3, x = 0$

Q3. $f''(x) = 6x - 8 \quad f'(2) = 0 \quad f(-1) = 3$

$f'(x) = 3x^2 - 8x + c_1$

$0 = 3(2)^2 - 8(2) + c_1$

$\therefore c_1 = 4$

$\therefore f'(x) = 3x^2 - 8x + 4$

$f(x) = x^3 - 4x^2 + 4x + c_2$

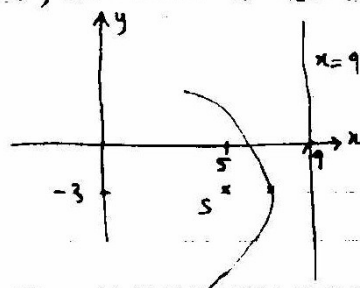
$3 = -1 - 4 - 4 + c_2$

$\therefore c_2 = 12$

$\therefore f(x) = x^3 - 4x^2 + 4x + 12$

23 (cont).

i) $S(5, -3)$, directrix is $x=9$.



$\therefore$ vertex is $(7, -3)$

$$a = 2$$

$$\therefore (y+3)^2 = -8(x-7)$$

$$y = \frac{1}{3}x^3 - x^2 - 8x - 6$$

y-intercept is -6

$$y' = x^2 - 2x - 8 = 0 \text{ for stat pts.}$$

$$(x-4)(x+2) = 0$$

$$\therefore x = 4, x = -2$$

$$y = -32\frac{2}{3}, y = 3\frac{1}{3}$$

$\therefore$ stat pts are $(4, -32\frac{2}{3})$ & $(-2, 3\frac{1}{3})$

$$y'' = 2x - 2$$

at $x=4$, $y'' = 6 > 0 \therefore$ minimum at $(4, -32\frac{2}{3})$

at $x=-2$, $y'' = -6 < 0 \therefore$ max at $(-2, 3\frac{1}{3})$

i) For pts of inflexion $y'' = 0$

$$\therefore 2x - 2 = 0$$

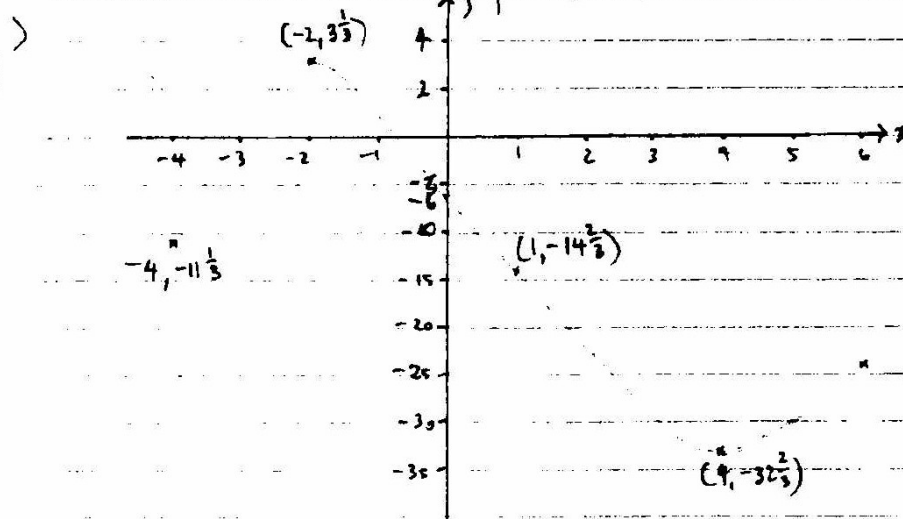
$$\therefore x = 1$$

$$y = -14\frac{2}{3}$$

x	0	1	2
y''	-2	0	+2

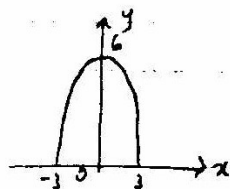
$\therefore$ change in concavity $\therefore (1, -14\frac{2}{3})$ is

a pt of inflexion



the minimum value is $-32\frac{2}{3}$

24) $y = 2\sqrt{9-x^2}$



Range: all real y where $0 \leq y \leq 6$.

1) $ax = \frac{3}{x-2}$

$ax(x-2) = 3$

$ax^2 - 2ax - 3 = 0$

For $y = ax$ to be a tangent to the hyperbola, there must only be one solution.

ie $\Delta = 0$.

$b^2 - 4ac = 0$

$4a^2 - 4(a)(-3) = 0$

$4a^2 + 12a = 0$

$4a(a+3) = 0$

$\therefore a = 0$ or $a = -3$.

2) $y = (\sqrt{x} + 1)^{100}$

$y' = 100(\sqrt{x} + 1)^{99} \times \frac{1}{2}x^{-\frac{1}{2}}$

$\therefore y' = \frac{50(\sqrt{x} + 1)^{99}}{\sqrt{x}}$

1) i) $f(x) = 2x^2 + x$

$f(x+h) = 2(x+h)^2 + (x+h)$

$\therefore f'(x) = \lim_{h \rightarrow 0} \frac{2(x+h)^2 + (x+h) - 2x^2 - x}{h}$

$= \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 + x + h - 2x^2 - x}{h}$

$= \lim_{h \rightarrow 0} \frac{h(4x + 2h + 1)}{h}$

$\therefore f'(x) = 4x + 1$

2) Show that $a(1 + f'(a)) = 2f(a)$

LHS = $a(1 + 4a + 1)$

$= 2a + 4a^2$

$= 2(2a^2 + a)$

$= 2f(a)$

$\therefore$ LHS = RHS.

c) Centre $(0,0)$ $r = 2$

$d = \frac{|5(0) - 12(0) + 26|}{\sqrt{5^2 + 12^2}}$

$d = \frac{26}{13} = 2$.

Since the perp. dist = the radius of the circle the line must be a tangent.

f) $f(x) = x^3 - 7x^2 + 1$

$f'(x) = 3x^2 - 14x$

$f''(x) = 6x - 14 < 0$ for concave down

$6x < 14$

$\therefore x < \frac{7}{3}$

Q5 a) $x^2 - 2x - 4 \geq 0$

$(x-4)(x+1) \geq 0$

$\therefore x = 4, x = -1$

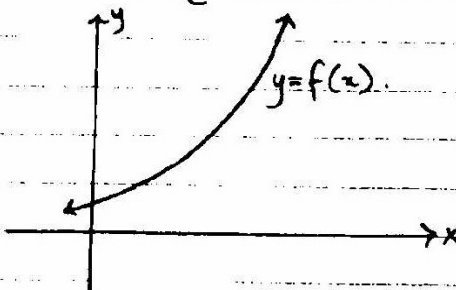


$\therefore x \leq -1, x \geq 4$

(2)

$\therefore$ Domain is: $\{x \in \mathbb{R} \text{ where } x \leq -1, x \geq 4\}$

b)



(2)

c) LHS = $\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$

$= \frac{1 - \tan^2 \theta}{\sec^2 \theta}$

$= \cos^2 \theta - \sin^2 \theta \times \frac{\cos^2 \theta}{1}$

$= \cos^2 \theta - \sin^2 \theta$

$= \text{RHS}$

$\therefore$ LHS = RHS QED!

(3)

25 (cont.) i)

$$SA = 2xh + 2yh + xy$$

$$V = xyh$$

$$V = 32$$

$$\therefore 32 = xyh$$

$$h = \frac{32}{xy}$$

$$SA = 2x\left(\frac{32}{xy}\right) + 2y\left(\frac{32}{xy}\right) + xy$$

$$SA = \frac{64}{y} + \frac{64}{x} + xy$$

$$\therefore SA = xy + 64\left(\frac{1}{x} + \frac{1}{y}\right) \text{ QED!}$$

$$= xy + 64\left(x^{-1} + \frac{1}{y}\right) \quad (3)$$

$$1) SA' = y - 64x^{-2} \quad (\text{keeping } y \text{ constant})$$

$$= y - \frac{64}{x^2} = 0 \text{ for stat. pts.}$$

$$\therefore \frac{64}{x^2} = y$$

$$x^2 = \frac{64}{y}$$

$$\therefore x = \frac{8}{\sqrt{y}} \quad (\text{since } x > 0)$$

Test for a minimum.

$$SA'' = \frac{128}{x^3}$$

when $x = \frac{8}{\sqrt{y}}$, $SA'' > 0 \therefore$ minimum Surface Area.

$$x < \frac{8}{\sqrt{y}} \quad A' < 0$$

$$\text{OR } x = \frac{8}{\sqrt{y}} \quad A' = 0$$

$$x > \frac{8}{\sqrt{y}} \quad A' > 0$$

$$\therefore x = \frac{8}{\sqrt{y}} \text{ gives } \text{min}$$

$$\therefore SA = \frac{8}{\sqrt{y}} \cdot y + 64\left(\frac{\sqrt{y}}{8} + \frac{1}{y}\right)$$

$$= 8\sqrt{y} + 8\sqrt{y} + \frac{64}{y}$$

(3)

$$SA = \left(16\sqrt{y} + \frac{64}{y}\right) \text{ m}^2 \text{ is the minimum surface area.}$$